

Indian Statistical Institute

BSDS - Second Year

Exercise Set 3.

1. Define for $a, b > 0$, $N_n^{a,b}$ as the number of paths $\pi = \{0 = s_0, s_1, s_2, \dots, s_n = a\}$ with $|s_i - s_{i-1}| = 1$, (starting from $(0, 0)$ to (n, a)) such that $s_i < a$ and $s_i > -b$ for all $i = 1, 2, \dots, n-1$. Note a, b can be ∞ also.

a) Show that

$$N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}.$$

b) Find $N_n^{a,\infty}$ and $N_n^{a+b,\infty}$.

2. For $n \geq 1$, show that

a)

$$u_{2n} = (-1)^n \binom{-\frac{1}{2}}{n}$$

and hence show that the generating function $U(s) = \sum_{n=0}^{\infty} u_n s^n$ is given by

$$U(s) = \frac{1}{\sqrt{1-s^2}}.$$

b)

$$f_{2n} = \frac{1}{2n-1} u_{2n} = (-1)^{n-1} \binom{\frac{1}{2}}{n}$$

and hence show that the generating function $F(s) = \sum_{n=0}^{\infty} f_n s^n$ is given by

$$F(s) = 1 - \sqrt{1-s^2}.$$

3. For $n \geq 1$, show that

a)

$$\mathbb{P}(S_1 \geq 0, S_2 \geq 0, \dots, S_{2n} \geq 0) = u_{2n}.$$

b)

$$\mathbb{P}(S_1 \geq -1, \dots, S_{2n-1} \geq -1) = 2u_{2n}.$$

c)

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

4.

a) Show that the expected number of returns to the origin in n steps is $\sum_{k=0}^n u_k \sim \frac{2}{\pi} \sqrt{n}$ as $n \rightarrow \infty$.

b) For $b \neq 0$, show that

$$\mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_n \neq 0, S_n = b) = \frac{|b|}{n} \mathbb{P}(S_n = b).$$

Show that

$$\mathbb{P}(S_{2n} = 0) = \frac{1}{2n} \mathbb{E}(|S_{2n}|).$$

Show that, for $b \geq 0$,

$$\mathbb{P}(\max\{S_0, S_1, \dots, S_n\} = b) = \mathbb{P}(S_n = b) + \mathbb{P}(S_n = b + 1).$$

c) Let $M_n = \max\{0, S_1, \dots, S_n\}$. For any $k > 0$, show that

$$\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).$$

5. For a simple symmetric random walk, show that

$$\mathbb{P}(\text{Infinitely Many returns to Origin}) = 1$$

6*. We say the particle spends the time interval $(i, i + 1)$ above the x-axis if at least one of the time points i and $i + 1$ the value is larger than 0. Otherwise, we say that the particle stays below the x-axis. Define

$$L_{2n} = \sum_{i=1}^{2n} \mathbb{I}\{\text{the particle spends } (i-1, i) \text{ is above x-axis}\}.$$

For $n \geq 1$, show that

$$\mathbb{P}(L_{2n} = 2k) = u_{2k} u_{2n-2k}.$$